

PROFESSIONAL SUMMARY

Quantitative developer and researcher with a PhD from the University of Oxford and an MSc from ETH Zürich, bringing over a decade of experience designing and deploying production-grade quantitative systems, spanning from high-frequency trading algorithms and low-latency GUIs/TUIs to large-scale ML pipelines and data infrastructure. Deep expertise across the full stack of quant development: research, implementation, testing, and deployment. Passionate about clean, maintainable code and the kind of software engineering practices that make colleagues happy rather than haunted.

PROFESSIONAL EXPERIENCE

Freelance Senior Quantitative Researcher / Developer | XO Market, Remote

March 2026 – today

- Designed and built a real-time terminal interface for monitoring and managing market-making activity on prediction markets, including live position tracking, order/quote management and risk alerts (Next.js)
- Built a Golang market-making strategy for prediction markets without a reliable fair-value anchor, estimating true probabilities from order flow and using inventory-aware quoting to manage adverse selection and directional risk
- Implemented a Golang market-making strategy quoting symmetric limit orders around the prevailing price, with each fill immediately hedged on Polymarket to stay risk-neutral

Senior Quantitative Researcher / Developer | Dialectic AG, Zug

Jan – Mar 2026

- Built and productionized a funding rate forecasting system using ARIMA and XGBoost, incorporating time-series analysis, feature engineering, and model validation with Python (Scikit-learn, Darts)
- Conducted statistical analysis of fund performance drivers, applying hypothesis testing, regression analysis, and quantitative modeling in Python to decompose and explain APR variability
- Led the implementation of secure, containerized development environments for agent-based systems, building a sandboxed Docker infrastructure and establishing security best practices for safe code execution and reproducibility

Senior Quantitative Developer | Digital Wave Finance AG, Zug

Mar 2025 – Dec 2025

- Architected and implemented a low-latency data ingestion and aggregation backend for CeFi and DeFi market data using Python, enabling real-time data access for quantitative research and trading systems (FastAPI)
- Designed and optimized scalable data schemas, APIs, and data pipelines to support high-throughput analytics workflows, improving data accessibility and reliability for downstream quant research and modeling

Senior Quantitative Developer | Mathrix AG, Zug

Oct 2022 – Jan 2025

- Designed and developed a low-latency trading graphical user interface for crypto spot and derivatives (perpetuals, options) using Python, enabling real-time order execution, market visualization, and interaction for the trading desk (PySide/Qt)
- Engineered a real-time monitoring and trade management system for hedging workflows, improving execution tracking, risk monitoring, and operational efficiency for traders using Python (PySide/Qt)
- Maintained a high-frequency market-making system for live crypto markets, implementing alpha-generating strategies, low-latency execution logic, and real-time data processing using Python and Node.js
- Led internal software engineering initiatives, including a Clean Code working group; delivered technical presentations on code quality, system design, and best practices to improve maintainability of the codebase
- Conceptualized and developed a transaction cost-aware hedging framework, incorporating execution costs, and optimization techniques using JavaScript
- Performed correlation and statistical analysis on crypto asset returns and US equity indices, applying time-series analysis, data analysis, and quantitative research methods in Python

Senior Data Scientist | SIX Group, Zurich

Jan 2019 – Oct 2022

- Delivered a company-wide big data analytics platform on Microsoft Azure, enabling large-scale quantitative data processing and research workflows for Databricks using PySpark and Hadoop
- Initiated and implemented a CI/CD pipeline on Azure DevOps, improving deployment reliability, code quality, and reproducibility for production-grade data
- Developed a fraud detection system for the Swiss Stock Exchange, leveraging deep autoencoder neural networks on transaction-level market data using Python (Pandas, NumPy, PyTorch, TensorFlow)
- Built a customer churn prediction model for financial services clients using deep learning techniques, handling real-world customer datasets with Python and PyTorch, supporting data-driven decision-making
- Built NLP-based document scraping pipelines using AWS Textract and Rekognition, enabling automated extraction, processing and classification of structured and unstructured customer data

Full Stack Software Engineer | PwC Digital Experience Center, Zurich

May 2016 – Jul 2018

- Led full-stack development of two high-performance e-commerce platforms, from architecture and implementation to maintenance, applying scalable database design, API development, and backend engineering using PHP7 and MySQL (Symfony2)
- Developed a recommendation engine for a major Swiss e-commerce platform, integrating Google Cloud Vision API and building data-driven algorithms for content-based recommendations, highlighting feature extraction, model integration, and predictive analytics

Quantitative Developer | SwissQuant Group AG, Zurich

Apr 2011 – Jul 2014

- Developed portfolio replication and optimization software for a Fortune 500 insurer, leveraging convex optimization solvers (Mosek, Gurobi, COIN-CLP) and MATLAB to deliver scalable, production-ready quantitative solutions
- Built a fully automated stock-picking system in MATLAB using regression models, statistical filters, and quantitative strategies, integrating with MySQL for data management and backtesting workflows
- Implemented stress-testing models integrated into a touch-enabled client advisory tool for a national private bank, applying risk modeling and scenario analysis with Java and MySQL
- Delivered internal training seminars on convex optimization at SwissQuant, providing mentorship and knowledge transfer on quantitative finance modeling and algorithmic development

EDUCATION

PhD — Numerical Particle Trajectory Simulation 2012 – 2016

University of Oxford

Doctoral research on numerical particle trajectory simulation using Monte Carlo simulation. Published 3 peer-reviewed papers; presented at 2 international conferences.

MSc — Mechanical Engineering (5.6 / 6.0) 2008 – 2010

ETH Zürich (Swiss Federal Institute of Technology)

Focus: Machine learning & pattern recognition, dynamic programming & optimal control, robotics & mechatronics

Exchange — Machine Learning & Advanced Robotics Jan – May 2009

University of Toronto

BSc — Mechanical Engineering (5.5 / 6.0) 2005 – 2008

ETH Zürich (Swiss Federal Institute of Technology)

Thesis: Control system for multi-agent coordination of micro-robots (C++)

PUBLICATIONS

- Grob, D. T., Wise, N., Oduwole, O., & Sheard, S. (2017). Magnetic susceptibility characterisation of superparamagnetic microspheres. *Journal of Magnetism and Magnetic Materials*, 452: 134–140.
- Oduwole, O., Grob, D. T., & Sheard, S. (2016). Comparison between simulation and experimentally observed interactions between two magnetic beads in a fluidic system. *Journal of Magnetism and Magnetic Materials*, 407: 8–12.
- Wise, N., Grob, D. T., Morten, K., Thompson, I., & Sheard, S. (2015). Magnetophoretic velocities of superparamagnetic particles, agglomerates and complexes. *Journal of Magnetism and Magnetic Materials*, 384: 328–334.

SKILLS & INTERESTS

Languages: German (native), English (full professional proficiency), French (fluent)

Technical: Python, MATLAB, JavaScript/Node.js, Rust (learning); PyTorch, TensorFlow, Scikit-Learn, Darts; FastAPI, PySide/Qt, PySpark; Azure, AWS; Docker, Git, CI/CD

Interests: Competitive athletics (Swiss Championship 800m & 1500m); tennis; climbing